Normal operators. We can extend Theorem 15 to the class of normal operators, which includes unitary and skew-adjoint operators. However, the eigenvalues are no longer guaranteed to be real.

THEOREM 16. *Let $\mathbb{V}$ be a complex, finite-dimensional inner product space, and let $T \in \mathcal{L}(\mathbb{V})$ be a normal operator. Then there exists an orthonormal basis of eigenvectors of T.*

Theorem 16 can be deduced from the following observation, which is an important result in itself.

PROPOSITION 7. *Let $\mathbb{V}$ be a finite-dimensional inner product space. Suppose $T, S \in \mathcal{L}(\mathbb{V})$ are self-adjoint and $TS = ST$. Prove that T (resp. S) is invariant on any eigenspace of S (resp. T). In particular, T and S can be diagonalized in a common basis.*

Variational characterization. We look at a variational characterization of eigenvalues, known as the min-max theorem, that allows us to define eigenvalues without having to associate eigenvectors.

THEOREM 17. *Let $\mathbb{V}$ be a finite-dimensional inner product space. Let T be self-adjoint and let $\lambda_1 \geqslant \lambda_2 \geqslant \cdots \geqslant \lambda_n$ be its eigenvalues. Then, for all $1 \leqslant k \leqslant n$,*

$$\lambda_k = \max_{\dim \mathbb{U} = k} \min \Big\{ \langle x, Tx \rangle : \ x \in \mathbb{U}, \ \|x\| = 1 \Big\},$$

where $\mathbb{U}$ denotes a subspace of $\mathbb{V}$. In particular,

$$\lambda_1 = \max_{x \in \mathbb{V}, \|x\|=1} \langle x, Tx \rangle \qquad \text{and} \qquad \lambda_n = \min_{x \in \mathbb{V}, \|x\|=1} \langle x, Tx \rangle.$$

EXERCISE 94. *Using Theorem 17, establish the perturbation result in Problem 19.*

Positive operators. We conclude this part by discussing a particular class of self-adjoint operators that come up in various applications.

DEFINITION 46. *A self-adjoint operator is said to be positive semidefinite (resp. positive definite) if the eigenvalues of T are nonnegative (resp. positive).*

EXAMPLE 14. *Following are some positive semidefinite (PSD) and positive definite (PD) operators.*

(1) *The zero operator is PSD, while the identity operator is PD.*
(2) *For any $q \in \mathbb{V}$, the operator $Tx = \langle x, q \rangle \, q$ is PSD.*
(3) *The Hessian matrix of a twice continuously differentiable convex function.*
(4) *The Gram and covariance matrices associated with a system of vectors.*

The following is a useful characterization of PSD and PD operators.

EXERCISE 95. *Let $T \in \mathcal{L}(\mathbb{V})$ be self-adjoint. Show that*

$$T \text{ is PSD} \iff \forall\, x \in \mathbb{V} : \ \langle x, Tx \rangle \geqslant 0,$$

and

$$T \text{ is PD} \iff \forall\, x \in \mathbb{V} \setminus \{0\} : \ \langle x, Tx \rangle > 0.$$

CHAPTER 10

SINGULAR VALUE DECOMPOSITION

Truncated SVD. Let $A \in \mathbb{R}^{m\times n}$. The question of diagonalization does not arise if $m \neq n$ (rectangular matrix). In fact, we have even seen examples of square matrices which cannot be diagonalized. The next best is the singular value decomposition (SVD) of A.

THEOREM 18. *Let $A \in \mathbb{R}^{m\times n}$ and $r = \operatorname{rank} A \geqslant 1$. Then there exists orthonormal vectors $v_1, \dots, v_r \in \mathbb{R}^n$, orthonormal vectors $u_1, \dots, u_r \in \mathbb{R}^m$, and $\sigma_1 \geqslant \cdots \geqslant \sigma_r > 0$, such that*

$$A = \sigma_1 u_1 v_1^\top + \cdots + \sigma_r u_r v_r^\top. \tag{15}$$

The existence of the decomposition in (15) can be deduced from the eigen decomposition of the Hermitian matrix $A^\top A$.

We call $\sigma_1, \dots, \sigma_r$ the singular values of A, $u_1, \dots, u_r$ are its left singular vectors, and $v_1, \dots, v_r$ the right singular vectors.

Full SVD. By extending the orthonormal vectors into orthonormal bases, we can write

$$A = U\Sigma V^\top, \tag{16}$$

where $U \in \mathbb{R}^{m\times m}$ and $V \in \mathbb{R}^{n\times n}$ are unitary matrices, and $\Sigma \in \mathbb{R}^{m\times n}$ is defined as

$$\Sigma_{ij} = \begin{cases} \sigma_i & 1 \leqslant i = j \leqslant r, \\ 0 & \text{otherwise.} \end{cases}$$

We call (15) the truncated (or partial) SVD of A and (16) the full SVD.

Relation with other concepts. The SVD and eigendecomposition of A need not coincide even if A is Hermitian. Indeed, while eigenvalues can take negative values, singular values are nonnegative by construction. However, they are the same if A is PSD.

EXERCISE 96. *Prove that for $A \in \mathbb{R}^{m\times n}$,*

$$\sigma_1(A) = \max\left\{\|Ax\|_{\mathbb{R}^m} : \|x\|_{\mathbb{R}^n} = 1\right\},$$

where $\|\cdot\|_m$ and $\|\cdot\|_{\mathbb{R}^n}$ are the Euclidean norms on $\mathbb{R}^m$ and $\mathbb{R}^n$.

In other words, $\sigma_1(A) = \|A\|_2$, the spectral norm of A, i.e., the matrix norm induced by the standard Euclidean norms on $\mathbb{R}^n$ and $\mathbb{R}^m$.

Consider the full SVD (16). Let $v_1, \dots, v_n \in \mathbb{R}^n$ be the columns of V and $u_1, \dots, u_m \in \mathbb{R}^m$ be the columns of U. The following result relates these to the four fundamental subspaces of A.

EXERCISE 97. *Show that $\mathcal{R}(A) = \operatorname{span}(u_1, \dots, u_r)$ and $\mathcal{N}(A) = \operatorname{span}(v_{r+1}, \dots, v_n)$.*

Similar to Theorem 17, we have the following min-max characterization.

EXERCISE 98. *Show that, for all* $1 \leqslant k \leqslant r$,

$$\sigma_k = \max_{\dim \mathbb{U}=k} \min \left\{ \|Ax\|_2 : \ x \in \mathbb{U}, \ \|x\|_2 = 1 \right\},$$

where $\mathbb{U}$ *is a subspace of* $\mathbb{V}$.

An important special case of the above result is that the largest singular value is simply the spectral norm of the matrix.

Applications. We have the following perturbation result as an application of the above min-max characterization.

EXERCISE 99. *Let* $A \in \mathbb{R}^{n\times n}$. *Let* $\operatorname{rank}(A) = r$ *and* $\sigma_1 \geqslant \cdots \geqslant \sigma_r$ *be the singular values of* A. *If* $\|E\|_2 < \sigma_r$, *prove that*

$$\operatorname{rank}(A + E) \geqslant r.$$

This means that a "small" perturbation cannot reduce the rank.

We can use the SVD to define the two important matrix norms—the spectral norm (largest singular value) and the nuclear norm (sum of the singular values).

EXERCISE 100. *Let* (15) *be the SVD of* $A \in \mathbb{R}^{m\times n}$. *Define* $\|A\|_2 = \sigma_1$ *and* $\|A\|_* = \sigma_1 + \cdots + \sigma_r$. *Verify that these are matrix norms on* $\mathbb{R}^{m\times n}$. *Moreover, prove that these are dual of each other in the sense that*

$$\|A\|_* = \max_{\|X\|_2 \leqslant 1} \operatorname{tr}(X^\top A),$$

and

$$\|A\|_2 = \max_{\|X\|_* \leqslant 1} \operatorname{tr}(X^\top A),$$

where the variable $X \in \mathbb{R}^{m\times n}$.

We can use SVD to find the Moore-Penrose pseudoinverse of A, denoted by $A^\dagger$. This is a generalization of the concept of matrix inverse. For any given A, it can be shown that there exists a unique matrix $A^\dagger$ (Moore-Penrose pseudoinverse of A) such that such that $AA^\dagger$ and $A^\dagger A$ are symmetric, and

$$AA^\dagger A = A \quad \text{and} \quad A^\dagger A A^\dagger = A^\dagger.$$

EXERCISE 101. *Let* (15) *be the partial SVD of* $A \in \mathbb{R}^{m\times n}$. *Verify that*

$$A^\dagger = \sigma_1^{-1} v_1 u_1^\top + \cdots + \sigma_r^{-1} v_r u_r^\top$$

satisfies the four defining properties of the Moore-Penrose pseudoinverse of A. *Moreover, for all* $A \in \mathbb{R}^{m\times n}$ *and* $b \in \mathbb{R}^m$,

(1) *Verify that* $x_0 = A^\dagger b$ *is a solution of the normal equation* $A^\top A x = A^\top b$.
(2) *If* x *is any solution of the normal equation, prove that* $x - x_0 \perp x_0$.
(3) *Of all solutions of the normal equation, prove that* x_0 *has the smallest* $\|\cdot\|_2$*-norm.*

CHAPTER 11

APPLICATIONS

1. Linear regression

Linear regression is a powerful tool widely employed in data analysis, machine learning, and scientific research. This uses a linear (affine) model to capture the relationship between the dependent and the independent variables. We will consider a simple yet instructive application of linear regression: fitting a polynomial using linear regression. In many real-world scenarios, the relationship between the variables is not linear. Polynomial regression can capture these nonlinear relationships by fitting a polynomial equation to the data.

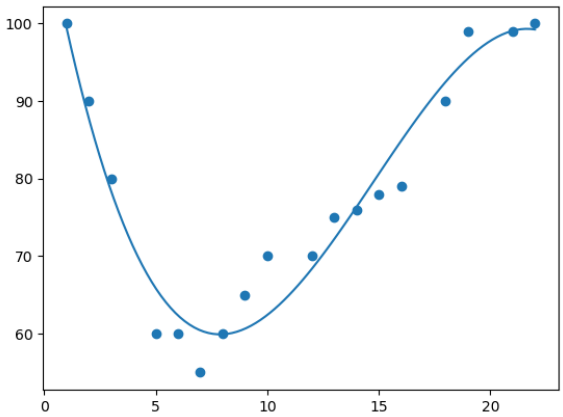

FIGURE 1. Fitting a cubic polynomial.

A polynomial of degree n can be expressed as

$$p(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \ldots + \beta_n x^n,$$

where y is the dependent variable, x is the independent variable, and $\beta_0, \ldots, \beta_n$ are the coefficients. Suppose we are given data points $(x_1, y_1), \ldots, (x_K, y_K)$. The fitting problem is to find a polynomial that best explains the data. In other words, we wish to find $\beta_0, \ldots, \beta_n$ such that $p(x_k) \approx y_k$ for all $k = 1, \ldots, K$. We use "$\approx$" since there might not exist a p that fits each point exactly (think of fitting three non-collinear points with a line). In least-squares regression, we wish to find $\beta_0, \ldots, \beta_n$ that minimizes

$$\ell(\beta_0, \ldots, \beta_n) = \sum_{k=1}^{K} (y_k - p(x_k))^2. \tag{17}$$

Despite the nonlinear nature of p, we can find the optimal coefficients that minimize (17) using linear algebra. More precisely, define the coefficient vector $\beta =$

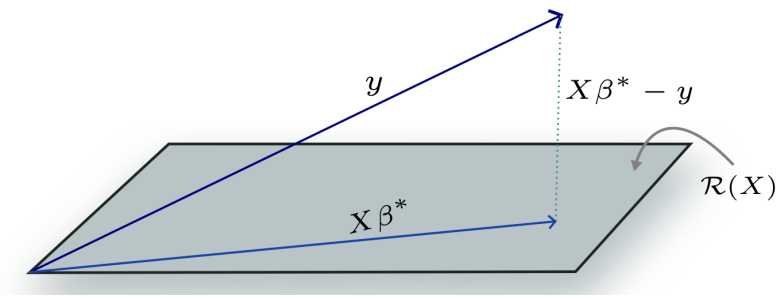

FIGURE 2. The normal equation can be written as $X^\top(X\beta^* - y) = 0$ which requires the residual vector $X\beta^* - y$ to be perpendicular to the range space of X. This comes essentially as a consequence of the Pythagoras theorem.

$(\beta_0, \ldots, \beta_n)$, the measurement vector $y = (y_1, \ldots, y_K)$, and the data matrix

$$X = \begin{pmatrix} 1 & x_1 & x_1^2 & \ldots & x_1^n \\ 1 & x_2 & x_2^2 & \ldots & x_2^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_K & x_K^2 & \ldots & x_K^n \end{pmatrix}.$$

We can then express the regression problem as

$$\min_{\beta \in \mathbb{R}^{n+1}} \ \ell(\beta) = \|y - X\beta\|^2, \tag{18}$$

where $\|\cdot\|$ is the standard Euclidean norm on $\mathbb{R}^K$. This is an instance of an unconstrained optimization where the loss function is quadratic. We wish to find β^* such that $\ell(\beta^*) \leqslant \ell(\beta)$ for all β; we call β^* a minimizer of (18). The following result connects this problem with the solution of linear systems of equations.

EXERCISE 102. *Prove that β^* is a minimizer of* (18) *if and only if $X^\top X\beta^* = X^\top y$.*

In other words, we can find the minimizer by solving $X^\top X\beta = X^\top y$. This is called the normal equation and has a nice geometric interpretation (see Fig. 2)

How do we know that the normal equation has a solution in the first place? We can use the range-nullspace decomposition of a linear transform to establish the following result.

EXERCISE 103. *Prove that, for any X and y, there exists β such that $X^\top X\beta = X^\top y$.*

In other words, we can always solve the regression problem (18) irrespective of the nature of the data points.

2. Optimization with linear constraints

We often encounter the problem of minimizing a differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ over an affine space $Ax = b$ ($A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$):

$$\min_{\beta \in \mathbb{R}^n} \ f(\beta) \quad \text{subject to } A\beta = b. \tag{19}$$

This is an example of a constrained optimization problem. Unlike linear regression, the variables in this case are restricted and cannot take on all possible values. However, note that the regression problem is a particular case of (19). Indeed, on setting $A = 0$ and $b = 0$ and $f(\beta) = \|X\beta - y\|^2$ in (19), we obtain (18).

We can characterize the solution of (19) using linear algebra.

EXERCISE 104. *Let $\beta^* \in \mathbb{R}^n$ be a minimizer of* (19). *Prove that there exists $\lambda \in \mathbb{R}^m$ s.t.*

$$\nabla f(\beta^*) = A^\top \lambda. \tag{20}$$

Note that, in general, the above condition is only necessary but not sufficient for β^* to be a minimizer of (19).

Next, consider the particular case

$$f(\beta) = \frac{1}{2}\beta^\top Q\beta, \tag{21}$$

where Q is a symmetric matrix. Then (20) becomes $Q\beta^* = A^\top\lambda$. Also, we have $A\beta^* = b$. This gives us the following system of $m + n$ equations in $m + n$ variables

$$\begin{pmatrix} A & 0 \\ Q & -A^\top \end{pmatrix} \begin{pmatrix} \beta^* \\ \lambda \end{pmatrix} = \begin{pmatrix} b \\ 0 \end{pmatrix}. \tag{22}$$

Assuming we can solve this system of equations, can we assert that β^* is the desired minimizer? Unfortunately, (20) is just a necessary condition, and the answer is generally no. However, if Q is positive semidefinite (f is convex), then β^* is indeed the desired minimizer.

EXERCISE 105. *Let β^* and λ be a solution of* (22) *and f be given by* (21) *where Q is PSD. Prove that β^* is the solution of* (19).

3. Principal component analysis

We consider the problem of approximating a set of points $x_1, \ldots, x_K$ in $\mathbb{R}^d$ by a subspace $\mathbb{V} \subset \mathbb{R}^d$. This comes up in dimensionality reduction techniques such as principal component analysis. We assume the points are centered around the origin.

The mathematical model for this approximation problem can be viewed as a generalization of linear regression to higher dimensions; however, there is a subtle difference in how the loss function is defined. More precisely, let $P_\mathbb{V} \in \mathcal{L}(\mathbb{R}^d)$ be the orthogonal projector onto the subspace $\mathbb{V}$. The "best" approximation of each x_k would be $P_\mathbb{V}(x_k)$ and the distortion incurred is the projection error $\|x_k - P_\mathbb{V} x_k\|$, where $\|\cdot\|$ is the standard Eulcidean norm. On summing the errors from all the points, we obtain the loss function

$$\ell(\mathbb{V}) = \frac{1}{K}\sum_{k=1}^{K} \|x_k - P_\mathbb{V} x_k\|^2, \tag{23}$$

where the variable is the subspace $\mathbb{V}$. Thus, finding the optimal subspace amounts to minimizing (23) w.r.t $\mathbb{V}$.

We can solve the above problem by representing $\mathbb{V}$ in an orthonormal basis and expressing (23) in terms of this basis vectors. More precisely, let $\dim \mathbb{V} = p$ and $\mathbb{V} = \operatorname{span}(v_1, \ldots, v_p)$, where $v_1, \ldots, v_p$ is an orthonormal basis of $\mathbb{V}$.

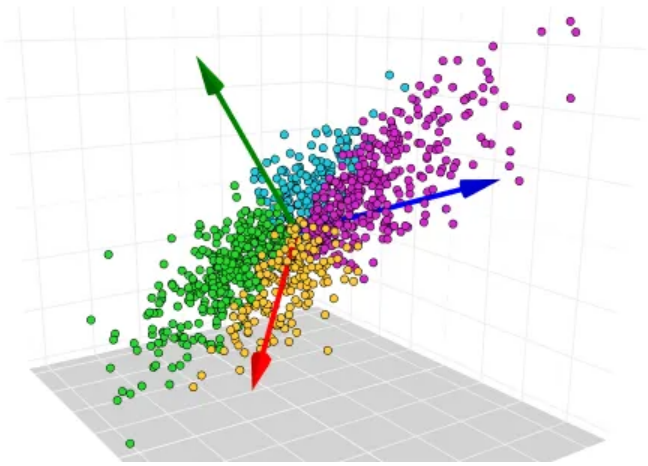

FIGURE 3. Approximating a point cloud in $\mathbb{R}^3$ by a plane. This is done by computing the eigenbasis of the covariance matrix.

EXERCISE 106. *For any $x \in \mathbb{R}^d$, show that*

$$\|x - P_{\mathbb{V}} x\|^2 = \|x\|^2 - \sum_{j=1}^{p} (x^\top v_j)^2. \tag{24}$$

In particular, define the covariance matrix

$$C = \frac{1}{K} \sum_{k=1}^{K} x_k x_k^\top.$$

Explain using (24) *why minimizing* (23) *is the same as maximizing*

$$\sum_{j=1}^{p} v_j^\top C v_j, \tag{25}$$

where the variables are the orthonormal vectors $v_1, \ldots, v_p$.

In other words, we have reduced the original subspace approximation problem to a routine eigenvalue problem. Notice that C is a PSD matrix with nonnegative eigenvalues.

EXERCISE 107. *Prove that* (25) *is maximized if we take $v_1, \ldots, v_p$ to be the top eigenvectors of C, i.e., the eigenvectors corresponding to the largest eigenvalues of C.*

A popular technique for reducing the number of features in a dataset without losing important information is the principal component analysis (PCA). This is modeled on the approximation problem discussed above. Based on the above analysis, we can summarize the key steps in PCA:

(1) Center the data to have zero mean.
(2) Compute the covariance matrix C of the centered data.
(3) Find the top-p eigenvectors of C (e.g. using the power method).
(4) Project the data onto the subspace spanned by these eigenvectors.

4. Low-rank approximation

Low-rank matrix approximation is a method in linear algebra and data analysis used to approximate a given matrix with a matrix of lower rank, often resulting in a more interpretable structure. The low-rank matrix acts as a compressed version of the original, saving space and computational time in various applications. This technique is useful in signal processing, machine learning, and data compression, as it reduces the complexity and dimensionality of data while retaining essential information.

Low-rank matrix approximation is closely connected to the singular value decomposition (SVD). According to the Eckart-Young theorem, the optimal rank-k approximation of a matrix is obtained by the k-term truncation of (15).

EXERCISE 108. *Let $A \in \mathbb{R}^{m\times n}$ be such that $r = \operatorname{rank}(A) \geqslant 1$. Consider the SVD of A given by* (15) *and, for $1 \leqslant k \leqslant r$, define*

$$A_k = \sigma_1 u_1 v_1^\top + \cdots + \sigma_k u_k v_k^\top.$$

Prove that

$$\|A_k - A\|_2 = \min_{\operatorname{rank}(X)\leqslant k} \|X - A\|_2,$$

where $\|\cdot\|_2$ is the spectral norm.

REMARK 18. *It turns out that the above result holds if we replace the spectral norm with the Frobenius or the nuclear norm. However, this need not be true for an arbitrary norm.*

5. Closest orthogonal transform

Identifying the closest orthogonal transform is vital in numerous scientific and engineering disciplines because it simplifies complex data and boosts computational efficiency while maintaining fundamental data properties. Applications include matching point clouds, analyzing shapes, estimating the motion of rigid objects, and comparing protein structures.

For example, suppose we have two points clouds $x_1, \ldots, x_n$ and $y_1, \ldots, y_n$ in $\mathbb{R}^3$. We wish to match x_i to y_i for $i = 1, \ldots, n$ using a rigid transform. In other words, we know the correspondence between the point sets (determining the correspondence is challenging in practice if it is not known beforehand), and we wish to "match" them using a rigid transformation. Since any rigid transform can be decomposed into an orthogonal transform and a translation, we can pose the matching problem in question as

$$\min_{O\in\mathbb{O}(3),\, t\in\mathbb{R}^3} \sum_{i=1}^{n} \|y_i - (Ox_i + t)\|_2^2,$$

where $\mathbb{O}(d)$ denotes the set orthogonal matrices acting on $\mathbb{R}^d$. On eliminating the translation t, we can reduce the above problem to that of finding $O \in \mathbb{O}(3)$ that minimizes $\|O - A\|_{\mathrm{F}}$ for some given A. Remarkably, we can find the closed-form solution using SVD.

EXERCISE 109. *Let $A \in \mathbb{R}^{d\times d}$ and let*

$$A = \sigma_1 u_1 v_1^\top + \cdots + \sigma_d u_d v_d^\top$$

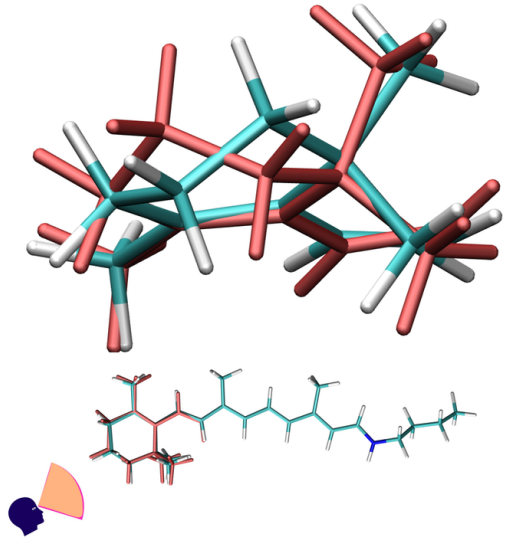

FIGURE 4. The task of aligning two protein sequences can be reduced to finding the closest orthogonal transform.

be its full SVD. Define

$$O^* = u_1 v_1^\top + \cdots + u_d v_d^\top.$$

Prove that

$$\|O^* - A\|_{\mathrm{F}} = \min_{O \in \mathbb{O}(d)} \|O - A\|_{\mathrm{F}}.$$

In other words, the closest orthogonal transform can be found by performing the SVD of the original transform and setting the singular values to 1.

6. Stationary distribution of a Markov chain

Consider a Markov chain on a finite state space of size n. This is described using a state transition matrix $P \in \mathbb{R}^{n \times n}$ having the properties:

$$\forall\, i, j = 1, \ldots, n: \; P_{ij} \geqslant 0; \quad \forall\, i = 1, \ldots, n: \; \sum_{j=1}^{n} P_{ij} = 1.$$

A probability distribution on the finite state space is described by a vector $\mu \in \mathbb{R}^n$, where $\mu_i \geqslant 0$ and $\mu_1 + \cdots + \mu_n = 1$. We will denote the collection of such probability vectors by Π_n. The Markov chain (as a random process) is completely determined by the initial distribution and the following transition rule: if the distribution at any epoch is μ, then the distribution in the next epoch is $\mu^\top P$.

We say that $\pi \in \Pi_n$ is a stationary distribution if $\pi^\top P = \pi^\top$. Thus, a stationary distribution is a probability distribution over the states that remain unchanged as the system evolves over time.

We call a Markov chain "regular" if there exists $k \geqslant 1$ such that $P^k > 0$. This is a special instance of a so-called irreducible Markov chain in which

$$\forall\, i, j = 1, \ldots, n, \; \exists\, k \geqslant 1: \quad (P^k)_{ij} > 0.$$